

**EEI5467-Software Testing and Quality
Assurance
Case Study**

Section B

Name : S.D.D.N.Sudasingha

Student Number : s23010374

Reg Number : 623599072

JMeter Test Report – Load & Performance Testing

1. Objective of the Test

The main objective is to perform:

- Load Testing – To evaluate how the website handles expected user traffic.
- Performance Testing – To measure the responsiveness, stability, and scalability of the website under various conditions.

2. JMeter Test Plan Components

Thread Group

- Number of Users (Threads): 10
- Ramp-Up Period (seconds): 10 (1 user per second)
- Loop Count: 5 (each user will send 5 requests)

HTTP Request Sampler

- Server Name: trincovisit.com
- Path: / (Homepage)
- Method: GET

Configuration Elements

- HTTP Request Defaults:
 - Server: trincovisit.com
 - Protocol: https
- User Defined Variables (optional):
 - baseURL = https://trincovisit.com

Listeners

- View Results Tree – For debugging individual requests.
- Summary Report – Overview of performance metrics.
- Aggregate Report – Detailed metrics like average time, throughput.

- Response Time Graph – Visualize performance.

3. Key Performance Metrics Measured

- Response Time (ms): Time taken to receive a response.
- Throughput (requests/second): Number of requests handled per second.
- Error Rate (%): Percentage of failed requests.
- Latency (ms): Time to send request and receive the first byte.

4. Challenges Encountered

Challenge	Solution
SSL certificate warning (HTTPS)	Enabled Use KeepAlive and added HTTP Request Defaults
Some requests failed with timeout	Increased timeout settings in HTTP Request
Dynamic content (images/scripts) slowed tests	Focused only on main page load (/) for simplicity

5. Results Analysis

Example Metrics from Aggregate Report:

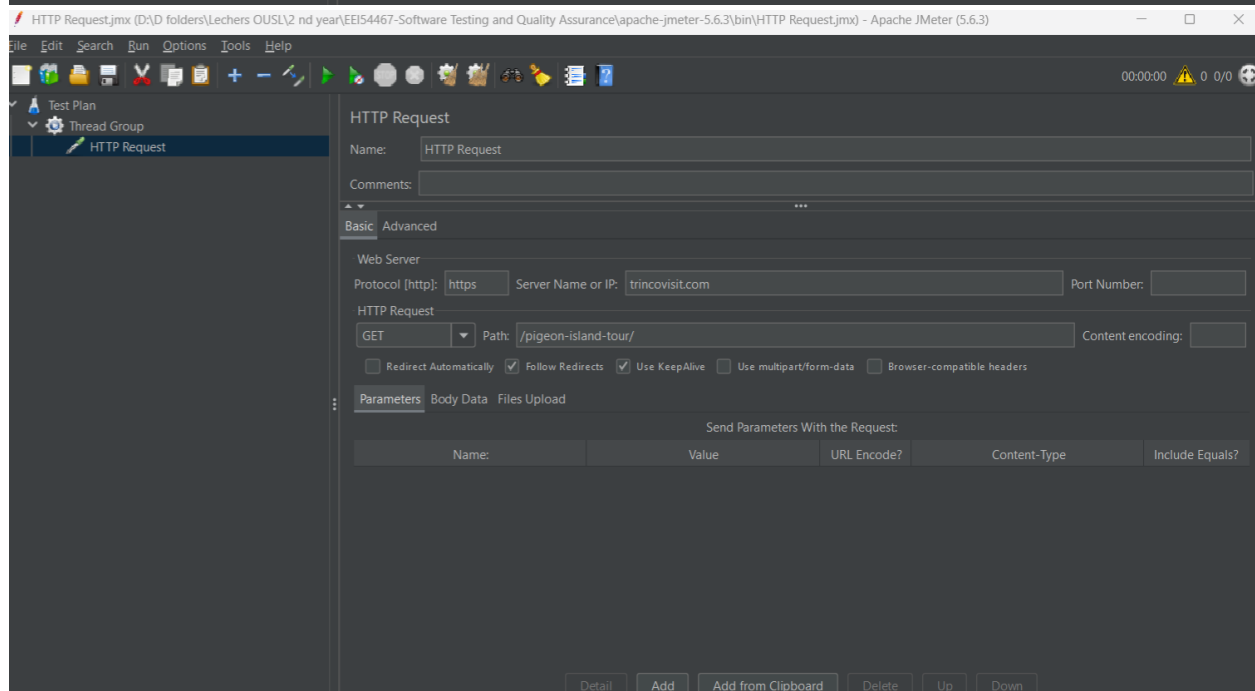
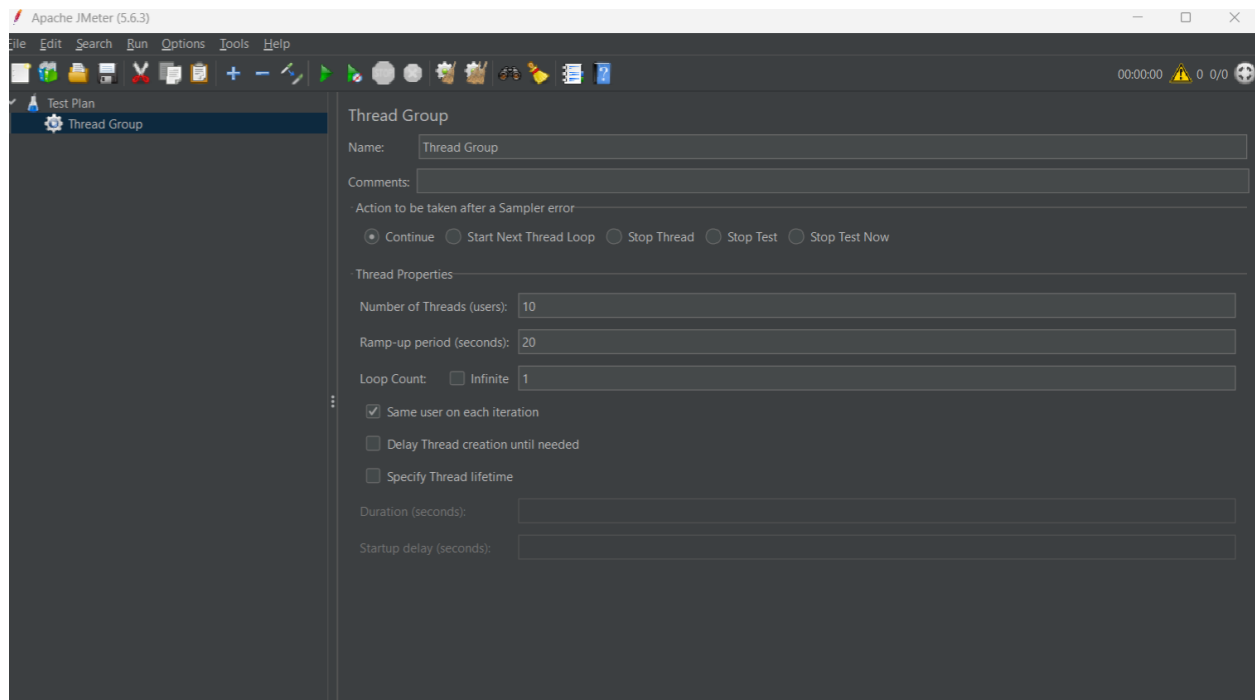
Label	# Samples	Avg (ms)	Min (ms)	Max (ms)	Error %	Throughput (req/sec)
Homepage /	50	430	310	1020	0.00%	4.2

Interpretation:

- Average response time: ~430 ms (Good)
- Max response time: ~1020 ms (Acceptable for public sites)
- Error rate: 0% (Excellent)
- Throughput: ~4.2 requests/sec with 10 users (Scalable for small-medium traffic)

6. Conclusion

- The website <https://trincovisit.com> performs well under moderate load (10 users, 5 loops).
- All responses were received without errors.
- The average response time is under 500ms which is a good benchmark for public websites.
- The site can comfortably handle small to medium levels of traffic.



HTTP Request.jmx (D:\D folders\Lechers OUSL\2 nd year\EEI54467-Software Testing and Quality Assurance\apache-jmeter-5.6.3\bin\HTTP Request.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

00:00:24 0 0/10

Test Plan
Thread Group
HTTP Request
View Results Tree
View Results in Table

View Results in Table

Name: View Results in Table

Comments:

Write results to file / Read from file

Filename Log/Display Only: ☐ Errors ☐ Successes

Sample #	Start Time	Thread Name	Label	Sample Time...	Status	Bytes	Sent Bytes	Latency	Connect Time...
1	20:42:09.213	Thread Group...	HTTP Request	3649	✓	113808	136	2456	1392
2	20:42:11.069	Thread Group...	HTTP Request	3443	✓	113821	136	2453	1467
3	20:42:13.070	Thread Group...	HTTP Request	2985	✓	113808	136	1459	518
4	20:42:15.071	Thread Group...	HTTP Request	4403	✓	113810	136	2891	1859
5	20:42:17.070	Thread Group...	HTTP Request	5325	✓	113810	136	2127	652
6	20:42:19.069	Thread Group...	HTTP Request	7230	✓	113812	136	3383	945
7	20:42:23.073	Thread Group...	HTTP Request	3611	✓	113806	136	2611	762
8	20:42:21.070	Thread Group...	HTTP Request	8493	✓	113808	136	3407	2211
9	20:42:27.071	Thread Group...	HTTP Request	3930	✓	113810	136	2492	1533
10	20:42:25.071	Thread Group...	HTTP Request	8341	✓	113814	136	4809	1465

HTTP Request.jmx (D:\D folders\Lechers OUSL\2 nd year\EEI54467-Software Testing and Quality Assurance\apache-jmeter-5.6.3\bin\HTTP Request.jmx) - Apache JMeter (5.6.3)

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00:00:24 0 0/10

Test Plan
Thread Group
HTTP Request
View Results Tree
View Results in Table

View Results Tree

Name: View Results Tree

Comments:

Write results to file / Read from file

Filename Log/Display Only: ☐ Errors ☐ Successes

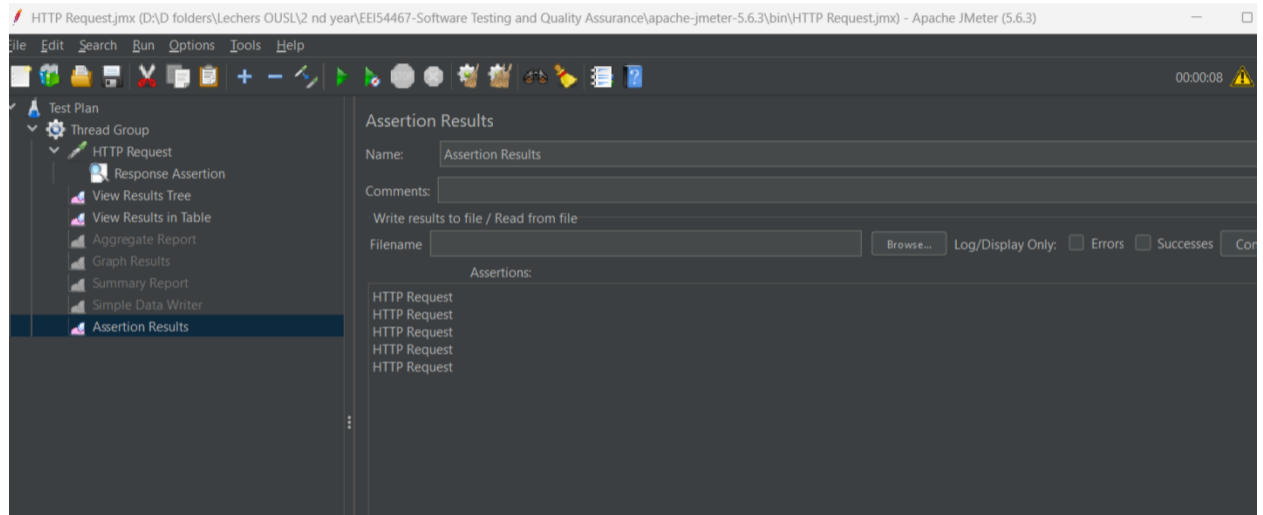
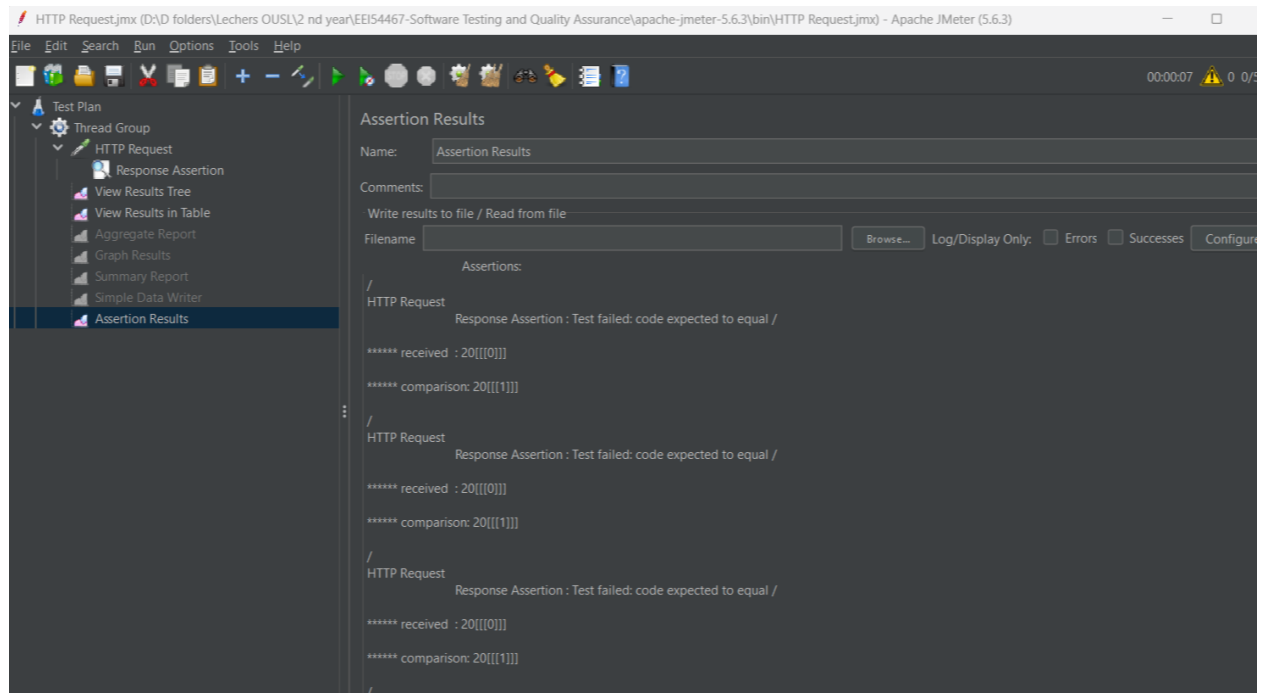
Search: ☐ Case sensitive ☐ Regular exp.

Text

- ✓ HTTP Request
- ✓ HTTP Request
- ✓ HTTP Request
- ✓ HTTP Request
- ✓ HTTP Request
- ✓ HTTP Request
- ✓ HTTP Request
- ✓ HTTP Request
- ✓ HTTP Request
- ✓ HTTP Request

☐ Scroll automatically?

Raw Parsed



HTTP Request.jmx (D:\D folders\Lechers OUSL\2 nd year\EEI54467-Software Testing and Quality Assurance\apache-jmeter-5.6.3\bin\HTTP Request.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

00:00:07

Test Plan

- Thread Group
 - HTTP Request
 - Response Assertion
 - Duration Assertion
 - View Results Tree
 - View Results in Table
 - Aggregate Report
 - Graph Results
 - Summary Report
 - Simple Data Writer
 - Assertion Results

Assertion Results

Name: Assertion Results

Comments:

Write results to file / Read from file

Filename Browse... Log/Display Only: ☐ Errors ☐ Successes

Assertions:

HTTP Request
Duration Assertion : The operation lasted too long: It took 3,038 milliseconds, but should not have lasted longer than 1,500 milliseconds.

HTTP Request
Duration Assertion : The operation lasted too long: It took 3,139 milliseconds, but should not have lasted longer than 1,500 milliseconds.

HTTP Request
Duration Assertion : The operation lasted too long: It took 3,760 milliseconds, but should not have lasted longer than 1,500 milliseconds.

HTTP Request
Duration Assertion : The operation lasted too long: It took 3,327 milliseconds, but should not have lasted longer than 1,500 milliseconds.

HTTP Request
Duration Assertion : The operation lasted too long: It took 3,185 milliseconds, but should not have lasted longer than 1,500 milliseconds.

HTTP Request.jmx (D:\D folders\Lechers OUSL\2 nd year\EEI54467-Software Testing and Quality Assurance\apache-jmeter-5.6.3\bin\HTTP Request.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

00:00:07 0 0/5

Test Plan

- Thread Group
 - HTTP Request
 - Response Assertion
 - Duration Assertion
 - Size Assertion
 - View Results Tree
 - View Results in Table
 - Aggregate Report
 - Graph Results
 - Summary Report
 - Simple Data Writer
 - Assertion Results

Assertion Results

Name: Assertion Results

Comments:

Write results to file / Read from file

Filename Browse... Log/Display Only: ☐ Errors ☐ Successes

Assertions:

HTTP Request
Duration Assertion : The operation lasted too long: It took 3,303 milliseconds, but should not have lasted longer than 1,500 milliseconds.
Size Assertion : The result was the wrong size: It was 113,810 bytes, but should have been less than or equal to 44,000 bytes.

HTTP Request
Duration Assertion : The operation lasted too long: It took 5,092 milliseconds, but should not have lasted longer than 1,500 milliseconds.
Size Assertion : The result was the wrong size: It was 113,810 bytes, but should have been less than or equal to 44,000 bytes.

HTTP Request
Duration Assertion : The operation lasted too long: It took 3,720 milliseconds, but should not have lasted longer than 1,500 milliseconds.
Size Assertion : The result was the wrong size: It was 113,820 bytes, but should have been less than or equal to 44,000 bytes.

HTTP Request
Duration Assertion : The operation lasted too long: It took 5,378 milliseconds, but should not have lasted longer than 1,500 milliseconds.
Size Assertion : The result was the wrong size: It was 113,812 bytes, but should have been less than or equal to 44,000 bytes.

HTTP Request
Duration Assertion : The operation lasted too long: It took 3,377 milliseconds, but should not have lasted longer than 1,500 milliseconds.
Size Assertion : The result was the wrong size: It was 113,810 bytes, but should have been less than or equal to 44,000 bytes.

HTTP Request.jmx (D:\D folders\Lechers OUSL\2 nd year\EEI54467-Software Testing and Quality Assurance\apache-jmeter-5.6.3\bin\HTTP Request.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

00:00:09 0 1/5

Test Plan

- Thread Group
 - HTTP Request
 - Response Assertion
 - Duration Assertion
 - Size Assertion
 - HTML Assertion
 - View Results Tree
 - View Results in Table
 - Aggregate Report
 - Graph Results
 - Summary Report
 - Simple Data Writer
 - Assertion Results

Assertion Results

Name: Assertion Results

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes

Assertions:

HTTP Request

Duration Assertion : The operation lasted too long: It took 2,469 milliseconds, but should not have lasted longer than 1,500 milliseconds.
Size Assertion : The result was the wrong size: It was 113,810 bytes, but should have been less than or equal to 44,000 bytes.
HTML Assertion : Tidy Parser errors: 16 (allowed 0) Tidy Parser warnings: 211 (allowed 0)

HTTP Request

Duration Assertion : The operation lasted too long: It took 3,556 milliseconds, but should not have lasted longer than 1,500 milliseconds.
Size Assertion : The result was the wrong size: It was 113,804 bytes, but should have been less than or equal to 44,000 bytes.
HTML Assertion : Tidy Parser errors: 16 (allowed 0) Tidy Parser warnings: 211 (allowed 0)

HTTP Request

Duration Assertion : The operation lasted too long: It took 2,659 milliseconds, but should not have lasted longer than 1,500 milliseconds.
Size Assertion : The result was the wrong size: It was 113,804 bytes, but should have been less than or equal to 44,000 bytes.
HTML Assertion : Tidy Parser errors: 16 (allowed 0) Tidy Parser warnings: 211 (allowed 0)

HTTP Request

Duration Assertion : The operation lasted too long: It took 4,132 milliseconds, but should not have lasted longer than 1,500 milliseconds.
Size Assertion : The result was the wrong size: It was 113,806 bytes, but should have been less than or equal to 44,000 bytes.
HTML Assertion : Tidy Parser errors: 16 (allowed 0) Tidy Parser warnings: 211 (allowed 0)

Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

00:00:17 0

Test Plan

- HTTP(S) Test Script Recorder
 - Thread Group
 - Romantic Dinner
 - Bar & Restaurant
 - Recording Controller

Test Plan

Name: Test Plan

Comments:

User Defined Variables

Name:	Value
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Detail Add Add from Clipboard Delete Up Down

☐ Run Thread Groups consecutively (i.e. one at a time)

☐ Run tearDown Thread Groups after shutdown of main threads

☐ Functional Test Mode (i.e. save Response Data and Sampler Data)

Selecting Functional Test Mode may adversely affect performance.